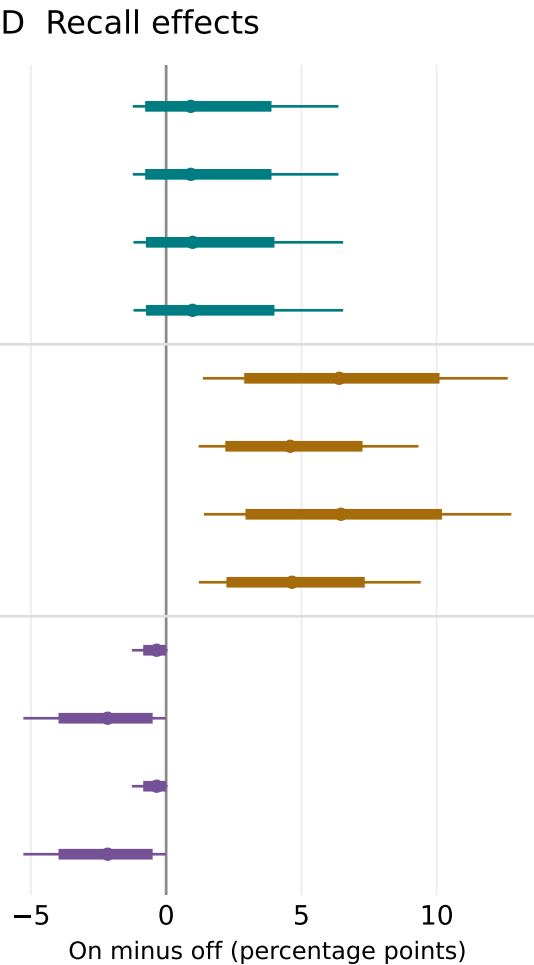
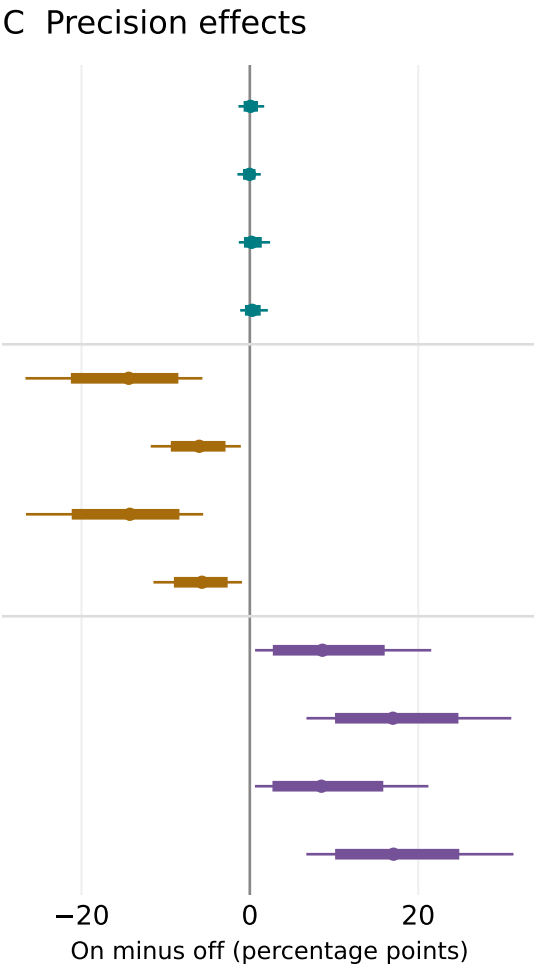
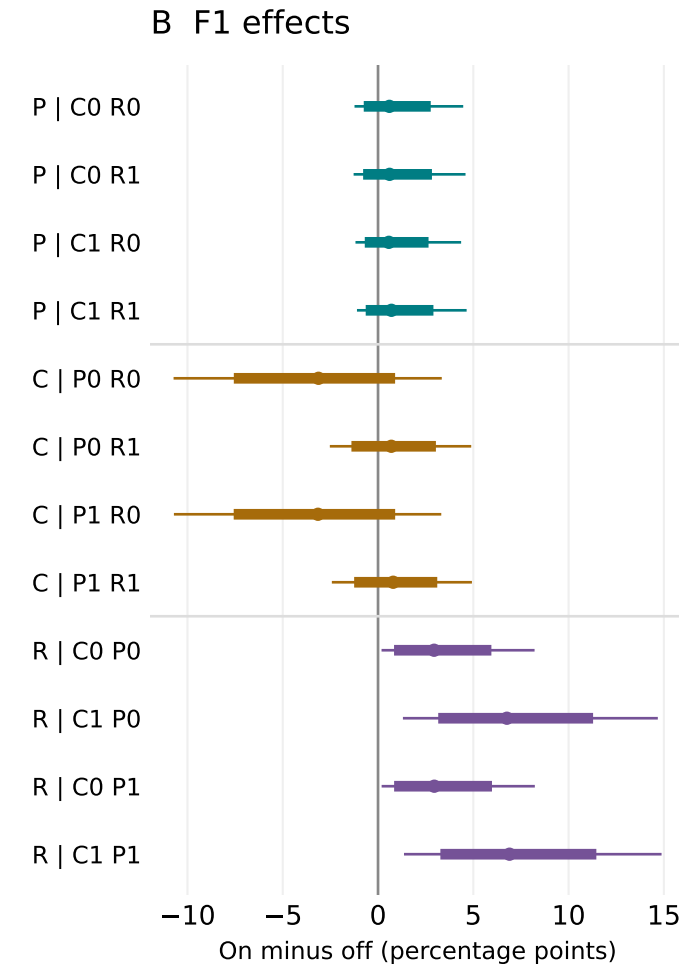
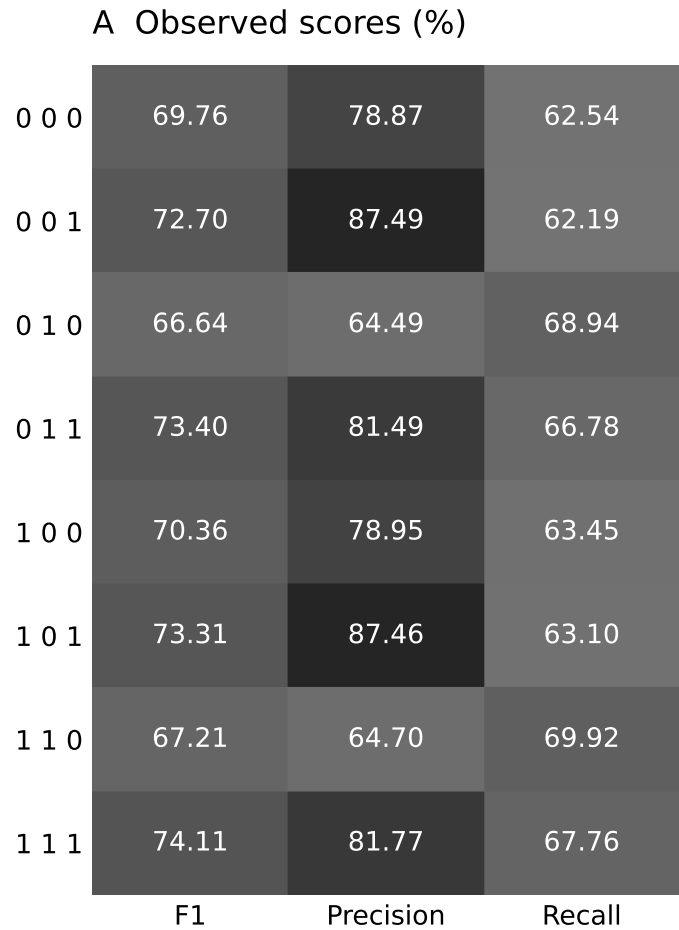


OrthoBench: HMM refinement, candidate expansion and reconciliation

70 reference families | 251,378 input genes | development-exposed component analysis

P = profile refinement; C = candidate expansion; R = reconciliation. Cell labels show P C R (0: off, 1: on).



Rows: profile effects (teal), candidate effects (ochre), reconciliation effects (purple); remaining factors held fixed.

Thick: nominal 95% CI. Thin: Bonferroni CI across all 36 endpoints. 20,000 paired RefOG draws; seed 20260918.

Profile-off retains initial HMM search. Reconciliation changes candidate co-membership to root HOGs. No independent superiority claim.